

Institute of Actuaries of India

Subject ST2 – Life Insurance

November 2008 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

- 1 (a) The bonus (r) is a proportion of the excess interest earned over the rate assumed in pricing and is expressed as a percentage of the reserves

If the premiums and the reserves increase by the same percentage, r , it is possible to increase the benefits by the same percentage

$$V \times (1+r) = SA \times (1+r) \times \bar{A} - P \times (1+r) \times \bar{a}$$

Non investment surplus can also be passed on to clients in the same manner or may be passed on to shareholders

A terminal bonus may be used to distribute capital appreciation

- (b) You need to invest carefully to fit the method of distribution:

- ? The bonus rate will be determined automatically, based on investment returns, so you need to ensure that returns are reasonably smooth from year to year (rules will normally define how returns are calculated)
- ? Money market investments tend to underperform compared to other types of investments, although they give a relatively smooth performance
- ? Use gilts, but long term gilts can lead to depreciation / appreciation which may produce volatile returns unless the rules exclude capital appreciation / depreciation)
- ? Corporate bonds can produce a higher yield than gilts and high quality bonds can be a suitable alternative although there is always some risk of default
- ? Very little in equity, as it is both volatile and produces a very poor income stream in the short term
- ? Fully developed property with secure tenants may be in order, as the income stream is fairly stable, and the prices are not normally as volatile as equities

[7]

- 2 (a) The key features of a unit linked contract are:

- ? The policyholder's premium is paid into an investment fund and at that time buys a number of "units", which represents a share of that fund.
- ? The value of the fund depends directly on the value of the assets underlying the investment fund.

- ? The value of one unit (the unit price) will be calculated daily as the value of the assets divided by the number of units
- ? The policyholder's share (ie the number of units) is unchanged until some policy cashflow occurs.
- ? The total value of an individual policyholder's fund at any time is the number of units allocated to the policy multiplied by the unit price
- ? The policyholder may have an option to switch the units to units of another investment fund of equivalent value
- ? The insurance company will deduct charges from the policyholder's fund. These may be deducted from the premiums before they are invested. For example:
 - o Less than 100% of premiums may be allocated to units
 - o There may be a bid offer spread
 - o A fixed amount may be deducted from each premium paid
- ? Charges may also be deducted from the unit funds. For example:
 - o A percentage of the fund value, taken on a regular basis
 - o A regular fixed charge
 - o Regular charges taken to cover risk benefits (eg life cover) in excess of the value of the unit funds.

These charges will be met by the insurer cancelling units of an equal value to the charges levied

- ? The contracts can be used to provide protection benefits in excess of the unit funds. The insurer needs to recoup the cost of these claims (plus a margin for profit) from the policy charges
- ? The value of the policy at maturity is usually the (bid) value of units
- ? There may be a guaranteed minimum maturity amount
- ? A surrender penalty may be deducted from the value of the units if the policyholder withdraws from the contract.

2 (b) Conditions for actuarial funding

- ? Charging structure that has higher (than "normal") levels of regular fund management charge on at least some of the units

- ? After actuarial funding the future profit flows projected on a prudent basis remain positive
- ? An appropriate unit related surrender penalty should be in place such that the surrender value cannot be greater than the actuarially funded unit reserve value at surrender
- ? There may be statutory constraints restricting the amount of actuarial funding that can be done in practice.
- ? Some accounting conventions also make no allowance for actuarial funding

2 (c) The changes to profit flows will be as follows:

- ? When a premium is allocated an increased profit flow will occur, because the cost of allocation (the amount of additional unit reserve required on premium allocation) has been reduced
- ? Profit during each subsequent year will be reduced because the fund management charge is based on a smaller unit reserve
- ? Profit each year will also be reduced by the cost of rebuilding the unit reserve gradually back towards the face value. The increase in cost will be spread over all the remaining years of the funding period
- ? On death, an additional cost equal to the face value less the funded value of the units will be incurred. The expected amount of this cost in any year will be equal to this difference multiplied by the appropriate mortality rate or may have been allowed for by using an appropriate endowment factor in calculating the actuarial funding.
- ? On surrender, a reduced profit will be made compared with the non funded position. The profit made on each surrender after actuarial funding, will equal the excess of the funded value of the units over the surrender value at the time of surrender.

[13]

3 (a) Suggestion to remove the suicide clause:

- ? In theory removal of any restrictive clause in the policy document would be welcomed by policyholders and may have some impact on the sales.
- ? However in practice it is difficult to see why this proposal would increase volumes sold, except to those likely to commit suicide.
- ? The merit of the suggestion should also be viewed in the context of the company's prevailing practice in allowing / disallowing claims due to suicide in the first year

- ? Life insurance companies usually find it difficult to disallow claims in the first year due to suicide for the fear of adverse publicity and hence any impact of removal is likely to be minimal.
- ? There will probably be a higher number of claims in the first year of the contract as some claims that would previously have been declined will be paid.
- ? The company may also need to decide whether the removal of the clause is with some retrospective effect to avoid any inconsistencies in the payment of claims.
- ? As the company has decided not to increase the premium rates, the extra cost of claims is then likely to reduce the profits than assumed when the product was priced.
- ? The company has to check whether the reduced profits still meets its profit criteria
- ? There will not be much data on which to base any assumption about the potential extra cost. Population statistics may be available.
- ? Reinsurance terms offered may deteriorate.
- ? People who are likely to value the removal of the clause are those who may be more likely to commit suicide. This may lead to anti-selection. The extent of this risk would depend on whether other companies still have a suicide exclusion clause.
- ? The company may think of introducing greater underwriting controls by seeking answers in the proposal form to establish any past suicidal tendencies. However this may be difficult to detect through normal underwriting.
- ? Any increase in underwriting could have counterproductive effect as these may act as barriers to sale

(b) The possible implications are:

- ? There will be less information on which to base the underwriting decision and so it will be more difficult to classify cases correctly into ordinary rates, special terms or decline.
- ? Hence it is possible that cases which otherwise might be rated or declined will be accepted.
- ? Alternatively a harsher approach might lead to cases that could be acceptable being declined.
- ? It will be necessary to monitor the effects of the change on profitability. In the early years there will only be limited data to do this, as there will be few deaths.
- ? If profitability reduces it may be necessary to increase premium rates and/or the valuation mortality assumption. The company may decide to take this step depending on the situation prevailing then
- ? There is a risk of anti-selection by those who can avoid disclosing relevant information.
- ? If there is reinsurance cover, this may need to be re-negotiated as the reinsurer may conclude that the anti-selection could materially impact the mortality experience.

- ? The changes may streamline the underwriting process, reducing costs, although there may be a greater need to request additional medical information, which would lead to additional costs.
- ? Simplifying the application may prove popular in the market, increasing new business, over which overheads can be spread.

(c) The common mortality options are:

- ? To purchase additional benefits without providing further evidence of health at the normal premium rates (for a life of that particular age) at the date on which the option is exercised
- ? To renew or lengthen the term of a life insurance policy eg a term assurance at the end of its original term without providing additional evidence of health
- ? To change part of the sum assured from one contract to another, eg from term assurance to endowment

(d) Conventional method for pricing mortality options:

- ? The method assumes that all eligible policyholders will take up the option
- ? The mortality of those taking the option is the same as the ultimate experience corresponding to the select rates used in the basic premium calculation
- ? If there is more than one option date, a “worst –case scenario” approach is adopted

(e) Calculation of additional single premium using North American Method:

The basic premium for original sum assured is:

$$P_0 = 10,00,000 \times \bar{A}_{[55]:5}^1 \quad \text{AM92 @4\%}$$

$$\begin{aligned}
 &= 10,00,000 \times (1.04)^{1/2} \times \left[\bar{A}_{[55]} - \frac{D_{60}}{D_{[55]}} \times A_{60} \right] \\
 &= 10,00,000 \times (1.04)^{1/2} \times \left[0.38879 - \frac{882.85}{1104.05} \times 0.45640 \right] \\
 &= 24,303.17
 \end{aligned}$$

The extra premium paid by the option takers at time 2

$$\begin{aligned}
 P_1 &= 5,00,000 \times \bar{A}_{[57]:5}^1 \\
 &= 5,00,000 \times (1.04)^{1/2} \times \left[\bar{A}_{[57]} - \frac{D_{60}}{D_{[57]}} \times A_{60} \right]
 \end{aligned}$$

$$= 5,00,000 \times (1.04)^{1/2} \times \left[0.41466 - \frac{882.85}{1010.66} \times 0.45640 \right]$$

$$= 8146.81$$

So at time 0:

$$\text{EPV of premiums} = 24,303.17 + \frac{D_{57}}{D_{[55]}} \times 0.3 \times 8146.81$$

$$= 24,303.17 + \frac{1012.34}{1104.05} \times 0.3 \times 8146.81$$

$$= 26,544.19$$

EPV of benefits =

$$10,00,000 \times \bar{A}_{[55]:\bar{a}}^1 + \frac{D_{57}}{D_{[55]}} \times \left[0.7 \times 10,00,000 \times \bar{A}_{57:\bar{a}}^1 + 0.3 \times 15,00,000 \times \left(\bar{A}_{57:\bar{a}}^1 \right)^* \right]$$

Where * denotes mortality at 250% of AM92 ultimate

The assurance terms in the above expression are:

$$\bar{A}_{[55]:\bar{a}}^1 = (1.04)^{1/2} \times \left[A_{[55]} - \frac{D_{57}}{D_{[55]}} \times A_{57} \right] = 1.0198039 \times \left[0.38879 - \frac{1012.34}{1104.05} \times 0.41556 \right]$$

$$= 0.0079027$$

$$\bar{A}_{57:\bar{a}}^1 = (1.04)^{1/2} \times \left[A_{57} - \frac{D_{60}}{D_{57}} \times A_{60} \right] = 1.0198039 \times \left[0.41556 - \frac{882.85}{1012.34} \times 0.4564 \right]$$

$$= 0.0178861$$

$$\left(\bar{A}_{57:\bar{a}}^1 \right)^* = 1.04^{-0.5} q_{57}^* + 1.04^{-1.5} (1 - q_{57}^*) q_{58}^* + 1.04^{-2.5} (1 - q_{57}^*) (1 - q_{58}^*) q_{59}^*$$

$$= 1.04^{-0.5} \times 2.5 \times q_{57} + 1.04^{-1.5} (1 - 2.5 \times q_{57}) \times 2.5 \times q_{58} + 1.04^{-2.5} (1 - 2.5 \times q_{57}) (1 - 2.5 \times q_{58}) \times 2.5 \times q_{59}$$

$$= 1.04^{-0.5} \times 2.5 \times 0.00565 + 1.04^{-1.5} (1 - 2.5 \times 0.00565) \times 2.5 \times 0.006352 + 1.04^{-2.5} (1 - 2.5 \times 0.00565) (1 - 2.5 \times 0.006352) \times 2.5 \times 0.00714$$

$$= 0.0138507 + 0.0147612 + 0.0157009$$

$$= 0.0443128$$

So:

EPV of benefits

$$=10,00,000 \times \frac{1012.34}{0.0079027 + \frac{1012.34}{1104.05}} \times [0.7 \times 10,00,000 \times 0.0178861 + 0.3 \times 15,00,000 \times 0.0443128]$$

$$=37,667.29$$

And the value of the option at time 0 is:

$$37,667.29 - 26,544.19 = 11,113.10$$

So the extra premium payable for the option is Rs 11,113.10

[27]

4 (a) The various risks which can have a material impact on a life company's solvency are:

1. The actual experience on the following turns out to be worse than expected / guaranteed,
 - ? investment returns
 - ? expenses
 - ? mortality/morbidity (or other contingency where relevant)
 - ? withdrawal rates.
2. New business mix deteriorates (sell more of the less profitable or capital intensive contracts).
3. New business volumes decline or explode.
4. New business strain causes solvency problems (this could be seen as a special case of expenses worse than expected and new business volume / mix worse than expected).
5. Discretionary payments to policyholders too large.
6. Surrender payments exceed asset share.
7. Creditors default (including reinsurer and asset defaults).
8. Supervisory authority imposes special restrictions or closes company to new business due to misconduct or insufficient financial strength.
9. Systems / staff unable to cope with new business.
10. Fraud, mismanagement.
11. Compensation payments need to be made for eg mis-selling
12. Legislation changes

Solutions to Risks identified in 1 above:

- ? Remove / reduce guarantees, including mortality, interest and expense guarantees (eg by writing variable charge unit-linked business).
- ? Reinsurance against excessive claims experience (risk premium and / or catastrophe).
- ? Underwrite to minimise the chance that policyholders' mortality is not worse than expected.
- ? Use stochastic asset / liability modelling to find appropriate risk / return balance for investment strategy, thereby maximising return for any given acceptable level of risk.

- ? Adapt investment policy to match assets and liabilities as closely as possible, including purchasing appropriate hedges
- ? Monitor experience as it develops, by specific investigations and by embedded value / profitability monitoring on continuously refined parameters. This will allow repricing or other action almost as soon as problems arise.
- ? Limiting chance of poor persistency through good quality sales process, good client service

Solutions to Risks identified under 2,3, 4 above include:

- ? Financial reinsurance against new business strain.
- ? Product design / pricing to minimise new business strain, maximise profitability and limit optionality.
- ? Attractive products, good bonus history, high commission to keep new business volumes
- ? Commission structure reflects desired new business mix.
- ? Actively monitor sales of capital intensive products

Solutions to Risks identified under 5 include:

- ? Discretionary payments managed to be close to smoothed asset share.

Solutions to Risks identified under 6 include:

- ? Surrender payments limited to smoothed asset share
- ? Surrender penalties on early withdrawal (and clawback of commission).

Solutions to Risks identified under 7

- ? Invest in diversified secure assets to protect against asset default.
- ? Deposit back of reinsurer balances

Solutions to Risks identified under 8 and 11 include

- ? Train sales staff and monitor their performance closely to reduce risks of mis-selling.
- ? Target solvency margin greater than minimum supervisory requirement to guard against deleterious random fluctuations.

Solutions to Risks identified under 9 include

- ? Design products with systems / staff in mind to protect against inadequacy.

Solutions to Risks identified under 10 include

- ? Protect against fraud by strict internal controls and checks.
- ? Protect against mismanagement by monitoring experience attributable to managers (eg new business volumes for marketing manager, embedded value growth for manager of a subsidiary).

Solutions to Risks identified under 12 include

? Don't write products which aim to exploit legal or fiscal anomalies

4 (b) Market consistent valuation of conventional without profits immediate annuities:

First project the net liability cash outgo that would be expected in each future year from the existing portfolio of policies. In this case, the cashflows will include annuity payments and expenses.

Assuming the cashflows are certain and guaranteed amounts, the market consistent value is then found by discounting the cashflows at current risk free rates of interest

Risk free rates may be defined as the rates on government securities or rates implied by swaptions

The risk free rate of interest to discount a guaranteed cashflow due in t years' time would be the current market redemption yield on a secure (probably government backed) zero coupon bond of the same term

Alternatively, determine the portfolio of such zero coupon bonds that could be owned at the present time that would produce exactly the same future cashflows as the liabilities.

The current market value of this portfolio is then the required market consistent valuation

This value may need adjustment for any short term anomalies in the market at the present time where the market is not deep and liquid

In reality the liability cashflows are not certain amounts, but are instead subject to uncertainties due to mortality, inflation, and any other factor that could affect the real level of expenses (including future volumes of in force policies)

Allowance may be made for the frictional cost of capital which is needed to back the liabilities – this may reflect the expenses of investing the capital and any tax impacts.

If we projected cashflows using best estimate assumptions for these variables the value we would obtain for the liabilities using the above approach would be lower than the market value

This is because a purchaser of the liability would require additional compensation for the possibility that the liability should turn out to be more expensive than expected

To allow for this uncertainty we would need to include “appropriate” margins in the expense, inflation and mortality assumptions compared with best estimate

Some market based indicators for some of these parameters might be available and would help determine the suitable level of margin to use

For example

- ? The difference between the current market yields of equivalent fixed interest and index linked bonds gives an indication of the market's expectation of future inflation
- ? There might be firms who provide third party administration services for life insurers, whose fees for taking on the administration of the portfolio would be the market level of expenses

[25]

5(a) All assumptions should reflect expected future experience in the new environment.

The company's recent expense experience over a period, long enough to pick up trends but recent enough still to be relevant would be analysed, allowing for inflation since the investigation. Expenses would be split between maintenance, claim and investment expenses.

Overheads previously allocated to new business expenses will need to be transferred to another category, although there might be some increment business that would take its share of overheads.

Exceptional costs as a result of the closure (e.g. redundancy costs) will need to be modelled over the period over which they are to be incurred.

There will be diseconomies of scale as the volume of in-force business declines and overheads have to be shared across fewer contracts. Therefore there is a need to project future expected expenses and in-force volumes of business, and hence there is a close link to the withdrawal assumption.

Allowance need to be made for potential shock lapses

The reasons for closure may have an impact

Analysis of expenses between fixed and variable would be useful, so that the latter can be modelled as per policy expenses in the embedded value calculation and varied by in-force volumes. The fixed expenses could be modelled separately, with an allowance for inflation.

Alternatively, the diseconomies of scale can be approximated by assuming a higher rate of expense inflation.

- (ii) Base on analysis of company's recent experience, split by duration, product and type of withdrawal and making allowance for special factors such as an adverse economic situation.

Allowance would be made separately for partial withdrawals and paid up rates

Consideration would be give to trends in persistency and the reasons for these trends where known

Allow for effect of closure to new business:

- If a sales force that is no longer employed serviced the clients, the level of contact will reduce and this may adversely affect persistency.
- If another life insurance company purchased the sales force then the clients are actively targeted and encouraged to transfer.
- Policyholders may have a lack of confidence in the company now that it has closed, which may cause surrenders to increase.
- There may be significantly higher withdrawals immediately following the closure that have not yet come through in the data
- However, if there are surplus assets in the fund then the prospect of sharing in their distribution may have a positive effect on persistency.

- (iii) The starting point is the assets currently backing the business, which are likely to be a mixture of equities/property and fixed interest.

The assumed yield on fixed interest will be based on the current average gross redemption yield with a reduction for risk depending on investment grade. Reinvestment may be assumed to be at lower rates.

The assumed return on equity and property is likely to be based on fixed interest yields plus a margin, where the margin may be derived from historical results.

The investment return on the with-profits fund is then taken as weighted average of these returns. As the fund declines and the portfolio matures, one could assume an increasing proportion invested in fixed interest. The speed at which this happens depends on the projected free estate of the company.

If strong, the fund can remain invested in real assets for longer, but the reasons for the closure may mean that a change in asset mix is necessary.

It is necessary for all the economic factors to be internally consistent, including the inflation assumption.

As there may be guarantees inherent in these products there is likely to be a stochastic valuation which will require further assumptions on eg volatilities, correlations and management actions.

[15]

- 6 The aim of catastrophe reinsurance is to reduce the potential loss to the ceding company due to any non-independence and accumulation of the risks insured.

It also lowers the probability of ruin in future scenarios that might be tested which are covered by the treaty, and avoids possible deterioration in the solvency position.

The cover is typically only available on a yearly basis and has to be renegotiated each year.

The reinsuring company will agree to pay out if a catastrophe, as defined in the reinsurance contract, occurs. There is no standard definition of a catastrophe but, typically, there needs to be a minimum number of deaths or sickness claims arising from a single incident occurring within a specified time of that incident.

The reinsurance contract will also specify how much the reinsurer will pay. This might be the excess of the total claim amount over the ceding company's catastrophe retention limit. The total claim amount would be net of any amounts reinsured on an original terms or risk premium basis.

The reinsuring company's liability in respect of a single catastrophe would be limited to a maximum amount. Any amount above that would fall back to the ceding company. There is also usually a maximum amount of cover per life.

The cover would usually exclude war risks, epidemics and nuclear risks.

[5]

- 7 If the NPV is calculated for an anticipated volume of business, then it provides the life insurance company with a realistic assessment of the likely profits that will arise over and above those required by the provider of capital.

The NPV is easy to calculate as it doesn't require an iterative approach.

The IRR method determines the rate of return expected to be achieved on capital invested and this could be compared to the RDR to assess whether

the return expected is satisfactory.

However, it doesn't provide information regarding the monetary size of the expected returns (over and above those required by the capital provider). The company may require a minimum monetary return for a product launch, for example, to be seen as worthwhile.

With the IRR method allowance for risk has to be made in the assumptions for the different elements of the cash flows.

The discounted payback period method calculates how long it takes the life insurance company to recover its initial investment with interest at the RDR.

But it doesn't provide information regarding the likely size of return that will arise over and above that required by the capital provider. In particular, the discounted payback period ignores the cashflows that arise once the capital has been repaid.

Using the NPV method, the life insurance company can relate the NPV to indicators of the policy's worth to the company, such as market share or sales effort. For example, the NPV can be expressed as a percentage of the premiums expected to be paid under a contract or the commission associated with selling that contract. This type of analysis isn't possible using the other methods.

There is only one net present value that can arise for a given profit signature and a given risk discount rate. For a given profit signature, the internal rate of return may not be unique (i.e. there may be more than one IRR), if there is more than one change of sign in the profit signature.

If a policy makes profits from outset or has a very low capital requirement, the DPP and IRR methods may not result in a meaningful number, or may not even exist. This is not true of the NPV method - an NPV always exists.

[8]
(Total 100 Marks)

*****END*****